# Cybersecurity Incident Report

| Section 1: Identify the type of attack that may have caused this network interruption |
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| The network interruption was caused by a **Denial-of-Service (DoS) attack**, specifically a **TCP SYN flood attack** targeting the company's web server (192.0.2.1). Packet analysis using Wireshark showed a large number of SYN requests coming from the unfamiliar IP address **203.0.113.0**. The attacker repeatedly sent SYN packets without completing the TCP three-way handshake, which caused the server to become overwhelmed with incomplete connection requests. Based on the log file, the IP Address that was flooding the web server is only IP Address that indicates DoS Attack and not DDoS Attack.<br><br>As the attack continued, the web server became unable to respond to legitimate employee traffic from the **198.51.100.0/24** network range. Employees attempting to access the website received **HTTP 504 Gateway Timeout** errors and connection reset messages, indicating the server was overloaded. |

| Section 2: Explain how the attack is causing the website to malfunction |
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| When a user connects to the website, the **TCP three-way handshake** is used to establish a connection. First, the client sends a **SYN** packet to request a connection. Second, the server responds with a **SYN-ACK** packet to acknowledge the request. Finally, the client sends an **ACK** packet to complete the handshake and establish the connection.<br><br>In this attack, the malicious actor sends a large number of **SYN packets** to the web server without completing the final step of the handshake that creates many incomplete connections.<br><br>The network logs show a high volume of SYN requests coming from the suspicious IP address **203.0.113.0**, while legitimate employee traffic originates from the **198.51.100.0/24** network range. As the number of SYN requests increases, the server becomes overwhelmed and cannot process legitimate requests.<br><br>As a result, employees attempting to access the website experience slow loading times and receive **HTTP 504 Gateway Timeout** errors or connection resets. This disrupts normal business operations and prevents employees from accessing the company's travel packages on the website. |

If this attack continues, it could lead to extended website downtime, loss of business opportunities, and reduced customer trust.

To prevent this type of attack in the future, the organization could implement security measures such as **intrusion detection systems, and firewall filtering**.